



MAHAMAYA POLYTECHNIC OF INFORMATION TECHNOLOGY

Department of Information Technology

MAJOR PROJECT REPORT

**“SecureSight AI: Criminal Detection and Surveillance System
Using Face Recognition”**

Submitted in partial fulfilment of the requirements for the award of
Diploma in Information Technology

Mr./Ms. _____

Date. _____

Bharat Shishodia – Team Leader, AI/ML & Backend Developer

Responsibilities:

Leading the complete project development process

- Project planning, task allocation, and team coordination
- Designing overall system architecture
- Developing backend using Python and Flask
- Implementing Face Detection and Face Recognition using OpenCV and LBPH
- Integrating AI modules with the web dashboard
- Managing criminal detection logic, evidence storage, and alert system
- Final testing, deployment, and project review
- Coordinating project documentation and presentation

Tarun jai – Database & Backend Developer

Responsibilities:

- Designing dataset structure and storage management
- Managing Authorized and Criminal datasets
- Implementing CRUD operations for criminal records
- Maintaining logs and evidence data
- Supporting backend integration and testing
- Assisting in model retraining and data management

Ayush Sharma – Frontend Developer

Responsibilities:

- Designing responsive user interfaces using HTML, CSS, and Bootstrap
- Developing Dashboard, Login, Evidence, and Criminal Database pages
- Implementing frontend navigation and user interactions
- Improving UI/UX design and responsiveness
- Integrating frontend pages with backend APIs

Aryan Sharma – Documentation & Testing

Responsibilities:

- Preparing project report and documentation
- Writing module descriptions and technical explanations
- Testing application functionality and performance
- Identifying bugs and reporting issues
- Preparing project presentation and viva support material
- Maintaining testing records and results

Ramu Yadav– Security Monitoring & Quality Assurance Engineer

Responsibilities:

- Monitoring system alerts and evidence management
- Verifying criminal detection outputs
- Performing system validation and quality checks
- Testing real-time camera and detection modules
- Assisting in deployment and troubleshooting
- Supporting final demonstration and project evaluation

Team Statement:

“This project is developed collaboratively by a team of five students specializing in **Artificial Intelligence**, **Backend Development**, **Frontend Development**, **Database Management**, **Testing**, and **Documentation**. The team worked together to design and develop an intelligent surveillance system capable of **detecting authorized persons**, identifying criminals, generating alerts, and maintaining digital evidence through **AI-powered face recognition technology**.”

Academic Session: **2025 – 2026**

Submitted To:

Department of Information Technology

Mahamaya Polytechnic of Information Technology, HATHRAS (U.P.)

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TEAM WORKFLOW DESCRIPTION

The SecureSight AI project was developed through effective teamwork and proper task distribution among all team members. Each member was assigned specific responsibilities based on their role, including AI/ML development, backend development, frontend design, database management, testing, and documentation.

Regular team meetings were conducted to monitor progress, discuss challenges, and ensure smooth integration of all modules. Through continuous collaboration and coordination, the team successfully developed a complete AI-based Criminal Detection and Surveillance System with real-time face recognition, alert generation, evidence management, and web-based monitoring capabilities.

CONTRIBUTION STATEMENT

Each team member made a significant contribution to the successful development of the SecureSight AI project. Responsibilities were distributed according to individual skills and areas of expertise, including AI/ML development, backend development, frontend design, database management, testing, and documentation.

The team worked collaboratively throughout the project lifecycle, ensuring effective communication, timely task completion, and smooth integration of all modules. Their collective efforts played a vital role in achieving the project objectives within the stipulated timeframe.

CERTIFICATE

This is to certify that the project entitled "**SecureSight AI: Criminal Detection and Surveillance System Using Face Recognition**" has been successfully completed by:

- Bharat Shishodia
- Tarun Jai
- Ayush Sharma
- Aryan Sharma
- Ramu Yadav

under the guidance of Shekhar Sir in partial fulfillment of the requirements for the award of Diploma in Information Technology during the academic session 2025-2026.

DECLARATION

We hereby declare that the project entitled “SecureSight AI: Criminal Detection and Surveillance System Using Face Recognition” submitted in partial fulfillment of the requirements for the award of Diploma in Information Technology is our original work.

The project has been developed by us under the guidance of the project supervisor. We further declare that this project has not been submitted to any other institution or university for the award of any degree, diploma, or certificate.

All sources of information used in this project have been properly acknowledged and referenced.

Date: _____

Place: _____

Project Team Members:

1. Bharat Shishodia (Team Leader)
2. Tarun Jai
3. Ayush Sharma
4. Aryan Sharma
5. Ramu Yadav

Signatures:

INTRODUCTION

SecureSight AI is an intelligent web-based surveillance and criminal detection system developed to enhance security through Artificial Intelligence and Face Recognition technology. The system is capable of detecting human faces in real-time, recognizing authorized individuals, identifying known criminals from a trained dataset, and generating security alerts when suspicious activity is detected.

The project is developed using Python, Flask, OpenCV, and LBPH Face Recognition algorithms. It provides a user-friendly web dashboard where administrators can monitor live camera feeds, manage criminal records, view evidence, and track security logs. The system automatically captures and stores evidence images whenever a criminal or unknown person is detected.

SecureSight AI aims to reduce manual surveillance efforts, improve security monitoring efficiency, and provide a smart automated solution for modern surveillance requirements.

PROJECT DETAILS

- Project Name: SecureSight AI – Criminal Detection and Surveillance System Using Face Recognition
 - Project Type: AI-Based Web Surveillance Application
 - Backend Technology: Python (Flask Framework)
 - Frontend Technology: HTML, CSS, Bootstrap, JavaScript
 - Artificial Intelligence Technology: OpenCV, LBPH Face Recognition
 - Database/Storage: Dataset-Based Storage and Log Management
 - Architecture: AI-Integrated Web Application Architecture
 - Project Category: Major Project
 - Academic Session: 2024 – 2025
-

PROBLEM STATEMENT

In traditional security systems, surveillance cameras continuously record footage, but monitoring these cameras manually is time-consuming, inefficient, and prone to human error. Security personnel may miss suspicious activities or fail to identify known criminals in real time.

There is a need for an intelligent surveillance system that can automatically detect faces, recognize authorized individuals, identify criminals from a predefined database, and generate instant alerts when suspicious persons are detected.

The SecureSight AI system addresses this problem by using Artificial Intelligence and Face Recognition technology to automate surveillance, improve security monitoring, store evidence, maintain logs, and provide real-time alerts through a user-friendly web dashboard.

OBJECTIVES

- To develop an intelligent surveillance system using Artificial Intelligence and Face Recognition technology.
- To detect and recognize human faces in real time using OpenCV.
- To identify authorized persons and known criminals from a trained facial dataset.
- To generate security alerts when a criminal or suspicious person is detected.
- To automatically capture and store evidence images for security purposes.
- To maintain activity logs for monitoring and investigation.
- To provide a user-friendly web dashboard for surveillance management.
- To reduce manual monitoring efforts and improve overall security efficiency.
- To integrate AI-based face recognition with a web-based application using Python and Flask.
- To create a scalable and cost-effective security solution for educational institutions, offices, and public places.

ACKNOWLEDGEMENT

We express our sincere gratitude to all those who have contributed directly or indirectly to the successful completion of our major project titled **“SecureSight AI: Criminal Detection and Surveillance System Using Face Recognition.”**

First and foremost, we would like to thank the Department of Information Technology, Mahamaya Polytechnic of Information Technology, for providing us with the opportunity to undertake this project as a part of our academic curriculum. This project has provided us with valuable practical knowledge and experience in the fields of Artificial Intelligence, Face Recognition, Web Development, and Security Systems.

We are deeply grateful to our project guide and faculty members for their continuous guidance, encouragement, and valuable suggestions throughout the development of this project. Their technical expertise and support helped us overcome various challenges and successfully complete the project.

We would also like to express our heartfelt thanks to our Principal, Head of Department, and all teaching staff members for providing the necessary resources, facilities, and motivation required for the successful execution of this project.

Special appreciation goes to all team members whose dedication, cooperation, hard work, and commitment played a significant role in the completion of this project. Through effective teamwork, regular discussions, and collaborative problem-solving, we were able to design and implement a fully functional AI-based surveillance system.

Finally, we would like to thank our friends, classmates, and family members for their constant encouragement, moral support, and motivation throughout the project development process.

This project has not only enhanced our technical knowledge but has also improved our teamwork, communication, problem-solving, and project management skills. We consider this project an important learning experience that will help us in our future academic and professional careers.

ABSTRACT

LBPH Algorithm, Evidence Management, Computer Vision. The rapid advancement of technology has increased the demand for intelligent security and surveillance systems capable of providing real-time monitoring and automated threat detection. Traditional surveillance systems mainly rely on continuous video recording and manual monitoring, which can be time-consuming, inefficient, and prone to human error. To address these challenges, the project **“SecureSight AI: Criminal Detection and Surveillance System Using Face Recognition”** has been developed as an AI-powered security solution.

SecureSight AI is an intelligent web-based surveillance system that utilizes Artificial Intelligence, Computer Vision, and Face Recognition technologies to identify authorized individuals, detect known criminals, and monitor suspicious activities in real time. The system captures live video streams through a webcam or surveillance camera, detects human faces using OpenCV, and recognizes individuals using the LBPH (Local Binary Pattern Histogram) Face Recognition algorithm.

The system maintains separate datasets for authorized persons and criminal records. Whenever a face is detected, it is compared with the trained dataset. If a match is found with an authorized person, the system grants access and displays the person's identity. If a criminal is detected, the system immediately generates an alert, activates a warning mechanism, stores photographic evidence, and records the event in system logs. Similarly, unknown individuals are identified and their images are saved for future investigation and monitoring.

A web-based dashboard developed using Flask, HTML, CSS, Bootstrap, and JavaScript provides administrators with a centralized platform for monitoring live camera feeds, viewing evidence, managing criminal records, accessing activity logs, and analyzing system statistics. The dashboard ensures easy operation and efficient management of surveillance activities.

The project demonstrates the practical implementation of Artificial Intelligence in modern security systems and highlights the potential of automated surveillance technologies in enhancing public safety. SecureSight AI not only reduces manual monitoring efforts but also improves accuracy, response time, and overall security effectiveness. The system can be further enhanced with advanced deep learning algorithms, cloud integration, multi-camera support,

mobile applications, and real-time notification services for large-scale deployment.

Keywords:

Artificial Intelligence, Face Recognition, OpenCV, Flask, Criminal Detection, Surveillance System, Security Monitoring,

Abstract - SecureSight AI is an intelligent surveillance platform developed using Python, Flask, OpenCV and LBPH Face Recognition. The system detects faces in real time, recognizes authorized users, identifies criminals from a predefined dataset, generates alerts, stores evidence images, and maintains logs through a web dashboard. This section explains the design, implementation, testing, deployment, and future enhancements of the project in detail.

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LITERATURE SURVEY

A literature survey is an essential part of any project as it helps in understanding the existing technologies, methodologies, and research work related to the proposed system. For the development of **SecureSight AI: Criminal Detection and Surveillance System Using Face Recognition**, various studies, research papers, and existing surveillance solutions were analyzed to identify their strengths, limitations, and possible improvements.

Traditional surveillance systems mainly depend on CCTV cameras and human operators for monitoring security activities. Although these systems are widely used in public places, educational institutions, offices, and transportation hubs, they require continuous manual observation. Human monitoring can be inefficient, especially when multiple cameras are involved, leading to delayed responses and missed security threats.

Recent advancements in Artificial Intelligence (AI) and Computer Vision have significantly improved the capabilities of surveillance systems. Face Recognition technology has emerged as one of the most effective biometric identification methods due to its accuracy, speed, and non-intrusive nature. Researchers have developed several face recognition techniques such as Eigenfaces, Fisherfaces, Local Binary Pattern Histogram (LBPH), and Deep Learning-based Convolutional Neural Networks (CNNs).

Among these methods, the LBPH (Local Binary Pattern Histogram) algorithm is widely recognized for its simplicity, efficiency, and effectiveness in real-time face recognition applications. It performs well under varying lighting conditions and requires relatively low computational resources, making it suitable for educational and small-scale security projects.

Several existing security systems provide face detection and recognition capabilities; however, many of them focus only on identification and lack features such as automatic evidence storage, criminal database management, real-time alert generation, and integrated web dashboards. Furthermore, commercial surveillance solutions are often expensive and require specialized hardware and software infrastructure.

The SecureSight AI project addresses these limitations by combining Face Detection, Face Recognition, Criminal Identification, Evidence Management, Activity Logging, and Web-Based Monitoring into a single integrated platform.

The system utilizes OpenCV for image processing, Flask for web application development, and the LBPH algorithm for facial recognition. The proposed solution aims to provide an affordable, scalable, and intelligent surveillance system capable of assisting security personnel in monitoring and identifying suspicious individuals in real time.

The survey of existing technologies and research works highlights the growing importance of AI-powered surveillance systems. It also demonstrates the need for intelligent security solutions that can automate monitoring processes, improve response times, reduce human effort, and enhance overall security management.

The findings from this literature survey served as the foundation for designing and developing the SecureSight AI system, ensuring that the proposed solution incorporates the strengths of existing technologies while addressing their limitations. Literature Survey - SecureSight AI is an intelligent surveillance platform developed using Python, Flask, OpenCV and LBPH Face Recognition. The system detects faces in real time, recognizes authorized users, identifies criminals from a predefined dataset, generates alerts, stores evidence images, and maintains logs through a web dashboard. This section explains the design, implementation, testing, deployment, and future enhancements of the project in detail.

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SCOPE OF PROJECT

The scope of the **SecureSight AI: Criminal Detection and Surveillance System Using Face Recognition** project is to provide an intelligent and automated security monitoring solution using Artificial Intelligence and Computer Vision technologies. The system is designed to detect, recognize, and monitor individuals in real time, thereby enhancing the efficiency and effectiveness of surveillance operations.

The project can be implemented in educational institutions, government offices, corporate organizations, shopping malls, hospitals, airports, railway stations, banks, and other public or private locations where security monitoring is essential. It helps security personnel identify authorized individuals, detect known criminals, and monitor unknown or suspicious persons without continuous manual observation.

The system provides several key functionalities, including real-time face detection, face recognition, criminal identification, evidence image storage, activity logging, and web-based surveillance monitoring. These features contribute to improved security management and faster response to potential threats.

SecureSight AI has the potential to be expanded in the future by integrating advanced technologies such as Deep Learning, Cloud Computing, Mobile Applications, Real-Time Notifications, GPS Tracking, and Multi-Camera Support. These enhancements can increase the scalability and effectiveness of the system for large-scale surveillance environments.

The project serves as a practical application of Artificial Intelligence in the field of security and demonstrates how modern technologies can be utilized to automate surveillance processes, reduce human effort, improve monitoring accuracy, and strengthen public safety measures.

Thus, the scope of SecureSight AI extends beyond academic learning and provides a foundation for developing advanced smart surveillance systems for real-world security applications.

Feasibility Study

Feasibility Study - SecureSight AI is an intelligent surveillance platform developed using Python, Flask, OpenCV and LBPH Face Recognition. The system detects faces in real time, recognizes authorized users, identifies criminals from a predefined dataset, generates alerts, stores evidence images, and maintains logs through a web dashboard. This section explains the design, implementation, testing, deployment, and future enhancements of the project in detail

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System Analysis

System Analysis - SecureSight AI is an intelligent surveillance platform developed using Python, Flask, OpenCV and LBPH Face Recognition. The system detects faces in real time, recognizes authorized users, identifies criminals from a predefined dataset, generates alerts, stores evidence images, and maintains logs through a web dashboard. This section explains the design, implementation, testing, deployment, and future enhancements of the project in detail. Detailed explanation paragraph 1.

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Requirement Analysis

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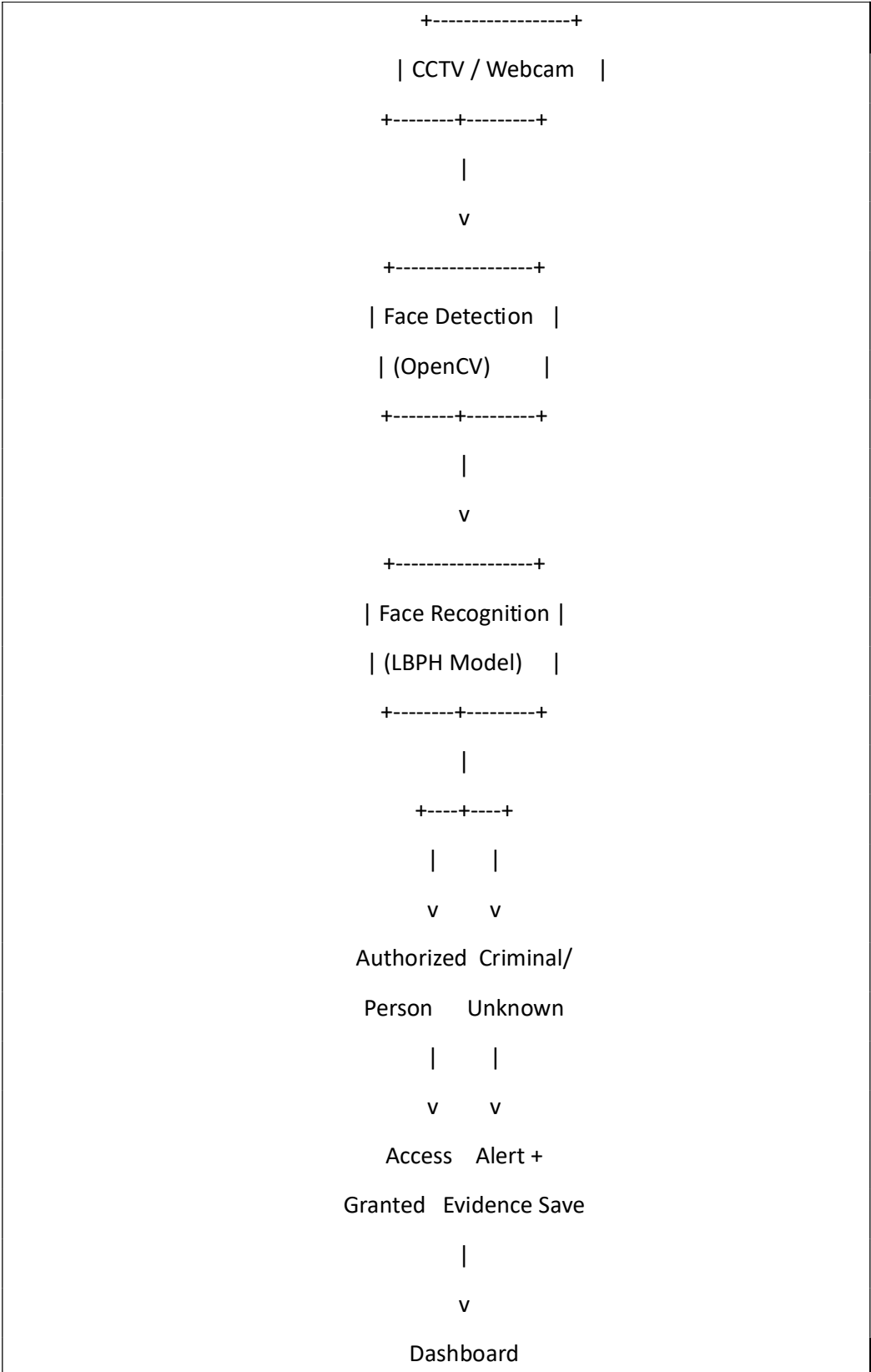
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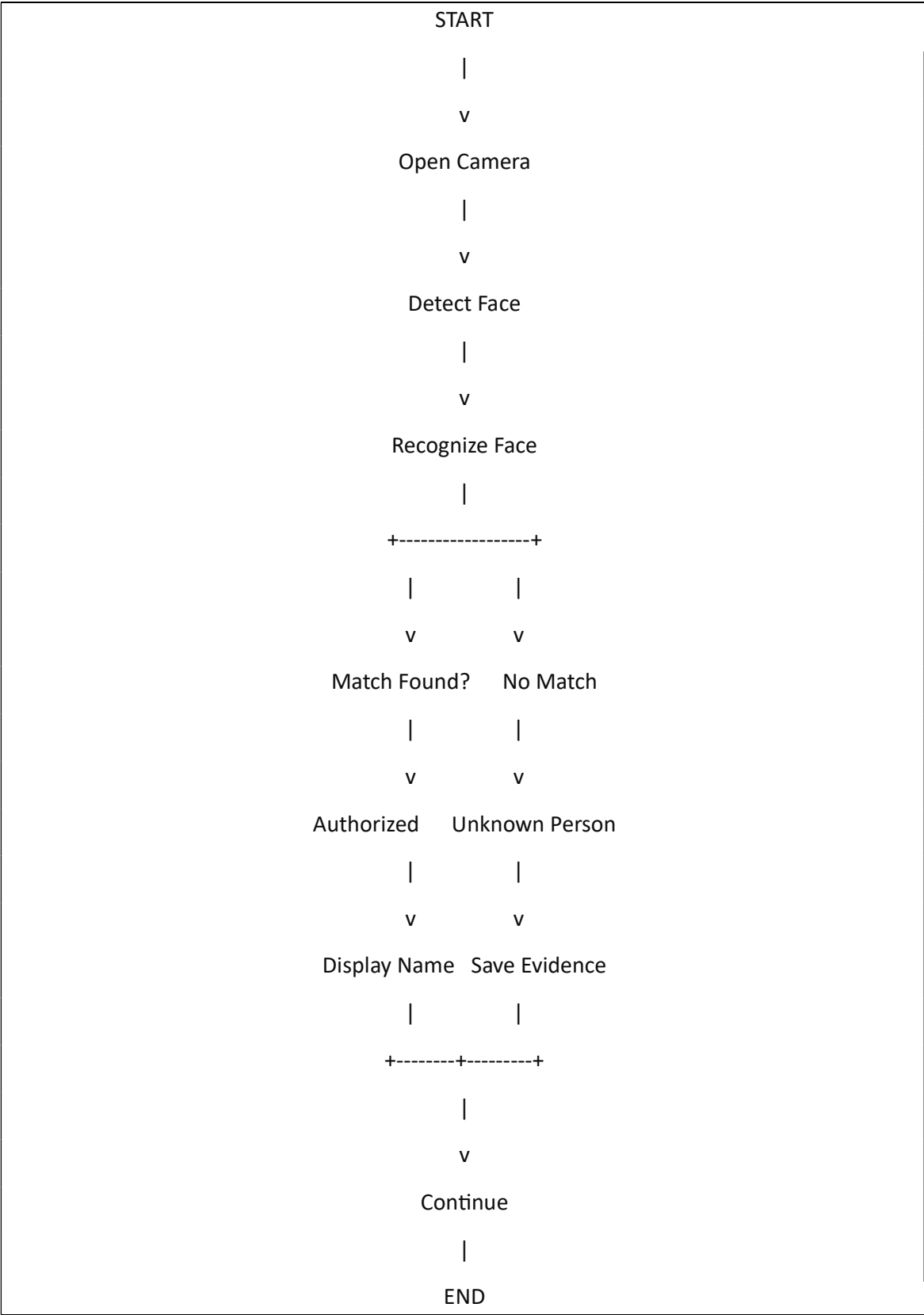
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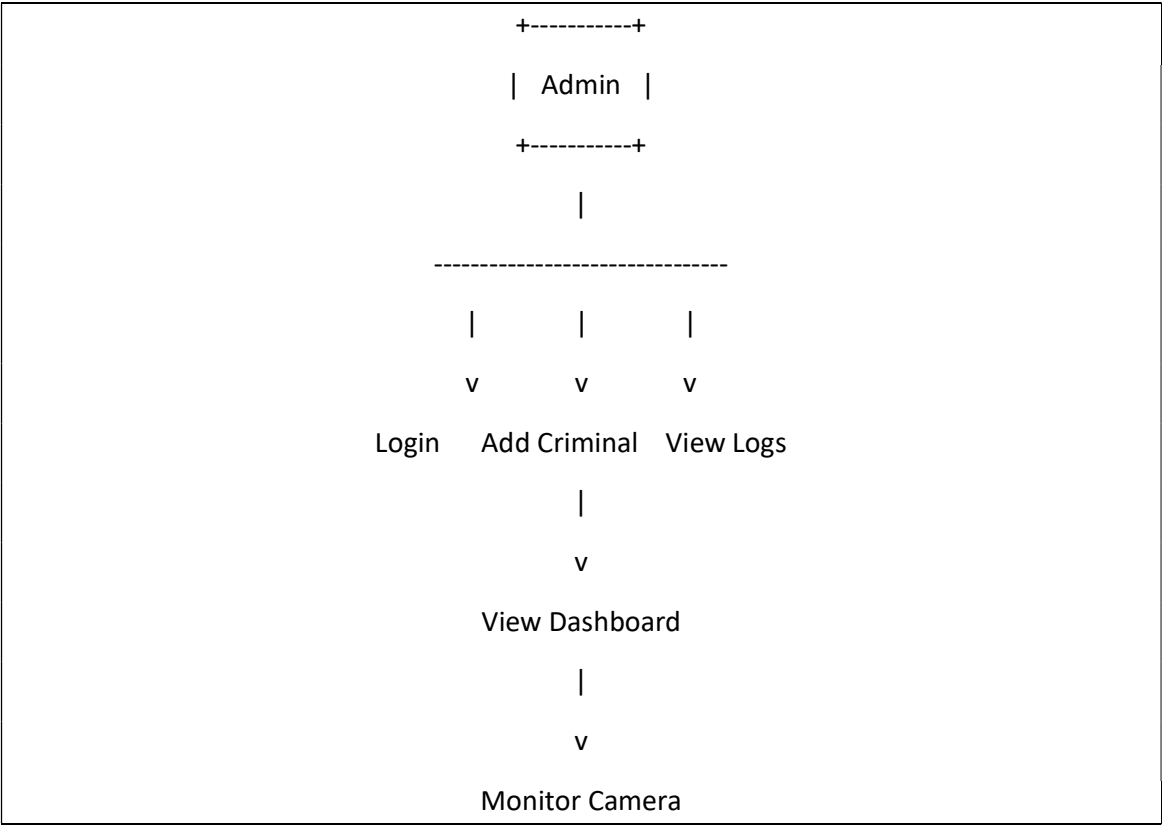
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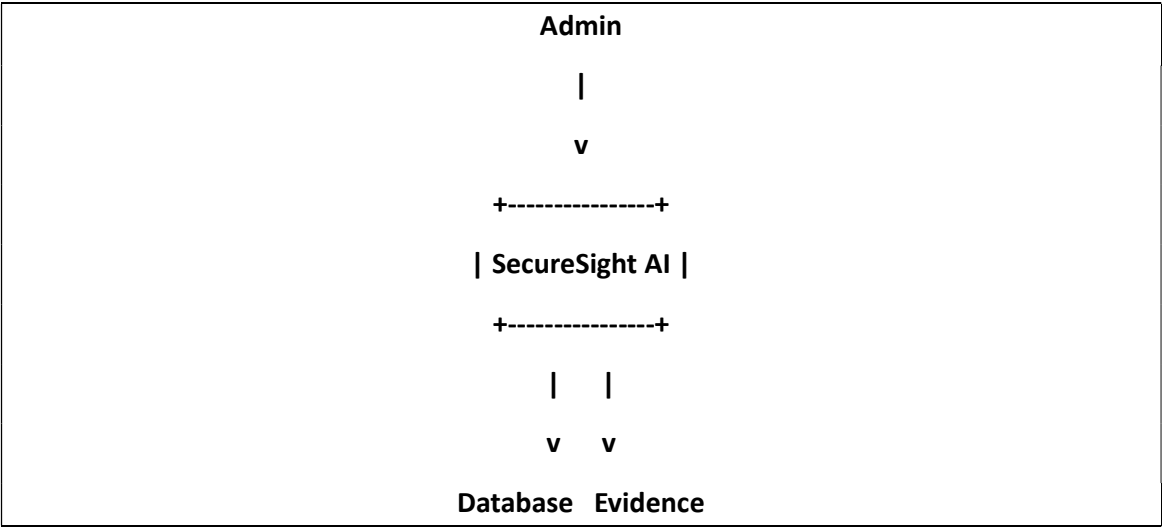
Flow Chart



Use Case Diagram



DFD Level 0

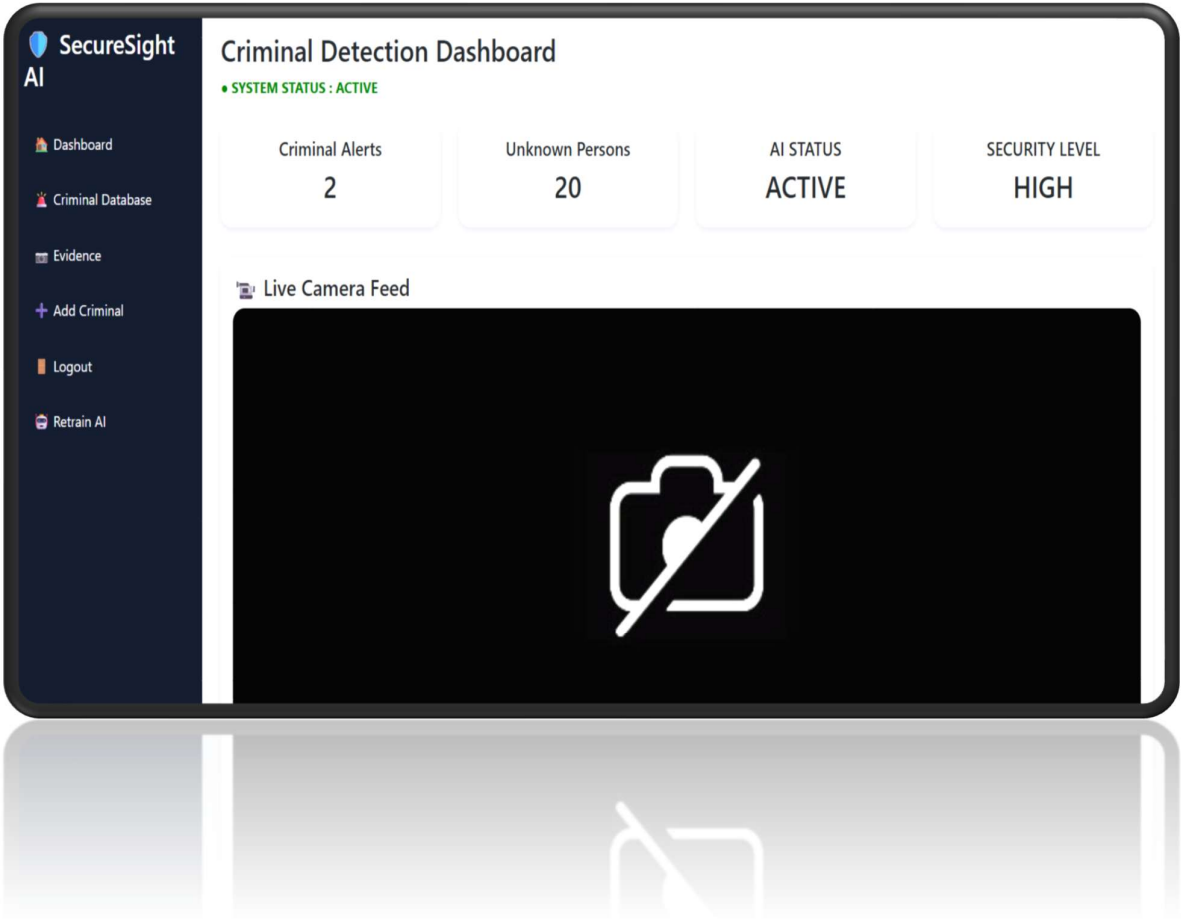


PROJECT ARCHITECTURE

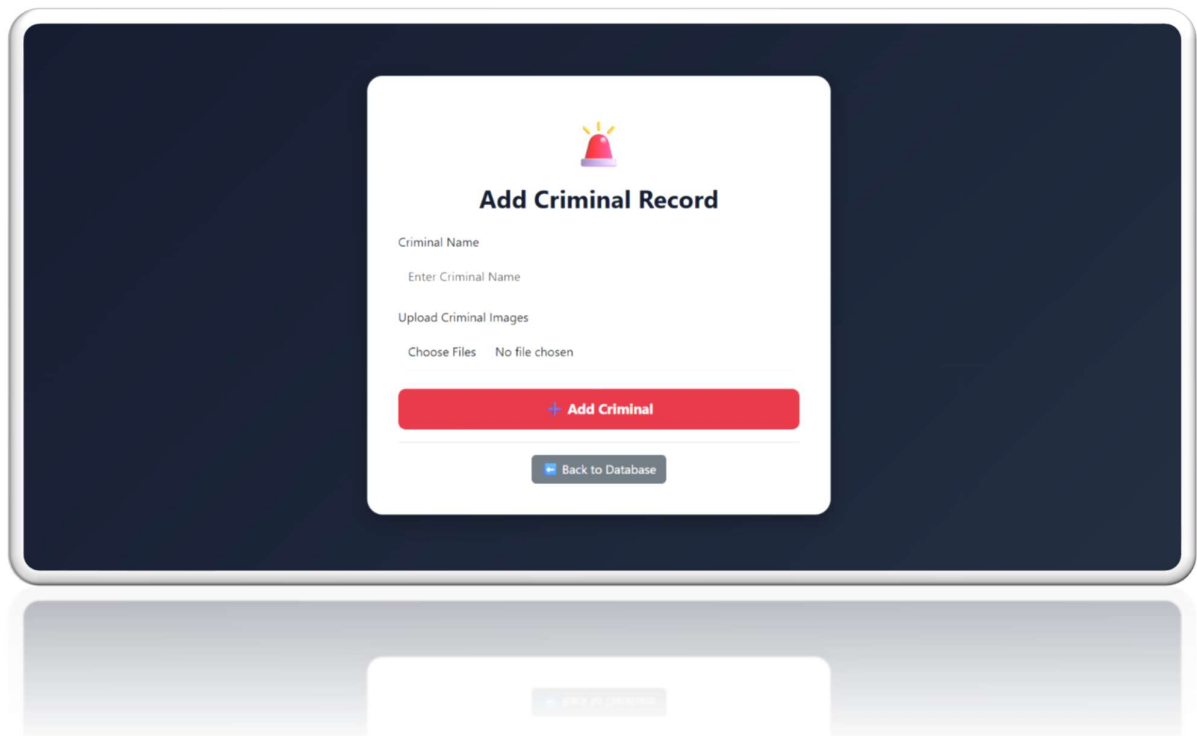
```
SecureSight AI
|
|└─ app.py
|
|└─ train_model.py
|
|└─ trainer.yml
|
|└─ siren.wav
|
|└─ authorized_dataset/
|  |└─ bharat/
|  |  |└─ 1.jpg
|  |  |└─ 2.jpg
|  |  └─ ...
|  |
|  └─ other_authorized_person/
|     |└─ 1.jpg
|     └─ ...
|
|└─ danger_dataset/
|  |└─ criminal1/
|  |  |└─ 1.jpg
|  |  |└─ 2.jpg
|  |  └─ ...
|  |
|  └─ criminal2/
|     |└─ 1.jpg
|     └─ ...
|
|└─ static/
|  |
|  |└─ css/
|  |  └─ style.css
|  |
|  |└─ js/
|  |  └─ script.js
|  |
```

```
| └─ evidence/
|   └─ alert/
|     └─ 171234567.jpg
|       └─ ...
|
| └─ suspicious/
|   └─ 171234890.jpg
|     └─ ...
|
| └─ templates/
|   └─ login.html
|   └─ dashboard.html
|   └─ evidence.html
|   └─ criminal_database.html
|   └─ add_criminal.html
|
| └─ logs/
|   └─ access_log.txt
|
| └─ uploads/
|   └─ temporary_files
|
| └─ venv/
|   └─ Python Virtual Environment
```


Project Screenshot: **Dashboard**



Project Screenshot: Add Criminal Record



The screenshot shows a web application interface for adding a criminal record. The main content is a white card centered on a dark blue background. At the top of the card is a red bell icon with yellow sound waves. Below the icon is the title "Add Criminal Record" in bold. The form contains three sections: "Criminal Name" with a text input field labeled "Enter Criminal Name"; "Upload Criminal Images" with a file upload area showing "Choose Files" and "No file chosen"; and a large red button labeled "+ Add Criminal". At the bottom of the card is a grey button labeled "Back to Database". The entire interface is displayed on a computer monitor with a reflection below it.

Add Criminal Record

Criminal Name
Enter Criminal Name

Upload Criminal Images
Choose Files No file chosen

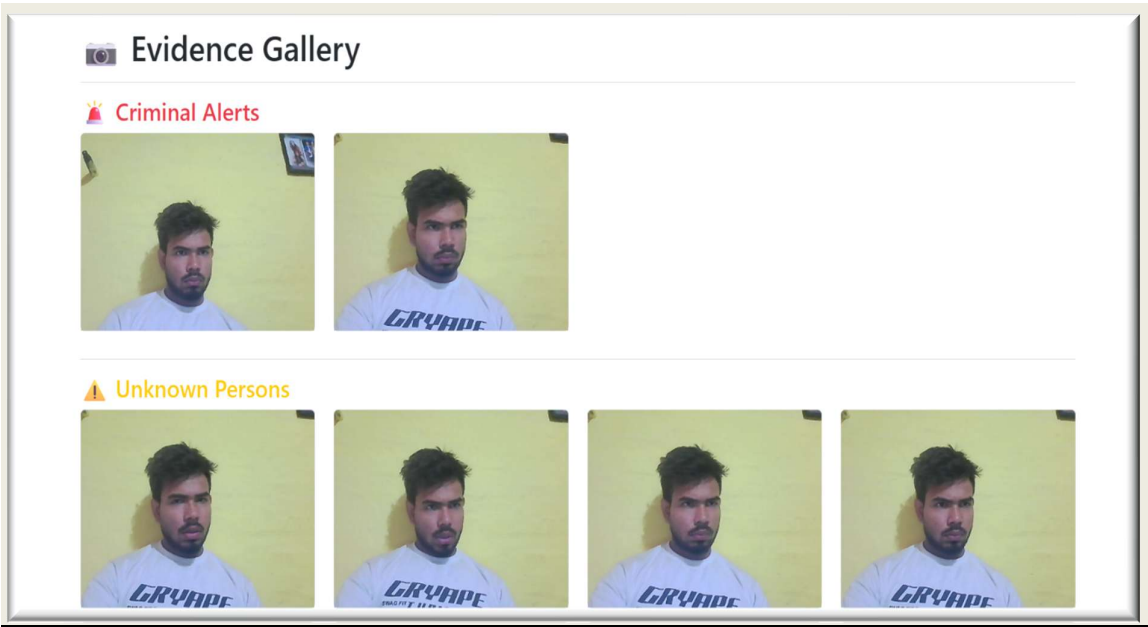
[+ Add Criminal](#)

[Back to Database](#)

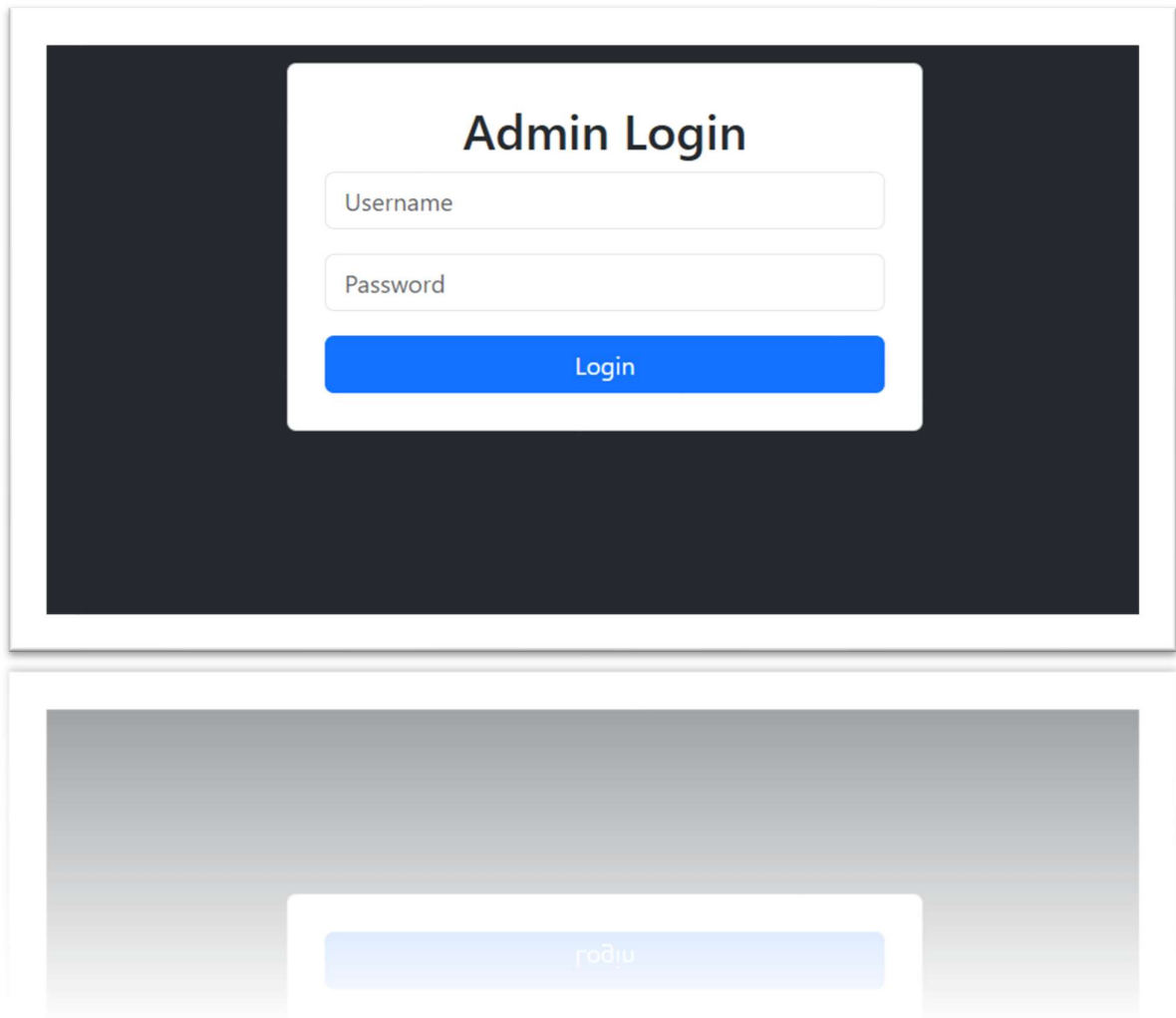
Project Screenshot : Criminal Database



Project Screenshot: Evidence Gallery



Project Screenshot: Admin Login



MODULE DESCRIPTION

1. Login Module

The Login Module provides secure access to the SecureSight AI system. Only authorized administrators can access the dashboard and system functionalities by entering valid login credentials. This module helps protect sensitive surveillance data from unauthorized access.

Functions:

- Admin authentication
 - Secure system access
 - Session management
 - Logout functionality
-

2. Dashboard Module

The Dashboard Module serves as the central control panel of the system. It displays live camera feeds, system status, criminal alerts, unknown person detections, and overall surveillance statistics in real time.

Functions:

- Live camera monitoring
 - System status display
 - Criminal alert statistics
 - Unknown person statistics
 - Navigation to different modules
-

3. Face Detection Module

This module uses OpenCV Haar Cascade Classifier to detect human faces from the live camera feed. It identifies facial regions and passes them to the recognition module for further processing.

Functions:

- Real-time face detection
- Face localization
- Image preprocessing
- Camera frame analysis

4. Face Recognition Module

The Face Recognition Module uses the LBPH (Local Binary Pattern Histogram) algorithm to recognize detected faces. It compares the captured face with the trained dataset and determines whether the person is authorized, criminal, or unknown.

Functions:

- Face recognition
- Dataset matching
- Identity verification
- Confidence score analysis

5. Criminal Database Module

This module maintains records of criminals stored in the danger dataset. Administrators can add new criminal records and update the dataset whenever required.

Functions:

- Criminal record management
- Image dataset storage
- New criminal registration
- Dataset organization

6. Evidence Management Module

Whenever a criminal or suspicious person is detected, the system automatically captures and stores evidence images. These images can later be used for investigation and verification purposes.

Functions:

- Automatic evidence capture
- Criminal evidence storage
- Suspicious person image storage
- Evidence retrieval

7. Alert System Module

The Alert System Module generates warnings whenever a criminal is detected. It activates the siren alarm and records the event in system logs for security monitoring.

Functions:

- Criminal alert generation
 - Siren activation
 - Emergency notification
 - Security event logging
-

8. Log Management Module

This module maintains a complete record of all system activities. Every criminal detection, unknown person detection, and important event is stored in log files for future reference.

Functions:

- Event recording
 - Activity tracking
 - Security auditing
 - Historical data management
-

9. Model Training Module

The Model Training Module trains the LBPH face recognition model using images stored in authorized and criminal datasets. After training, a trainer.yml file is generated for face recognition.

Functions:

- Dataset processing
 - Face extraction
 - Model training
 - Trainer file generation
-

10 Admin Management Module

This module allows the administrator to control the overall system, monitor activities, retrain the model, manage datasets, and supervise surveillance operations.

Functions:

- System administration
- Dataset management
- User monitoring
- Model retraining
- Security management

Advantages of Proposed System

- Real-time face recognition
 - Automatic criminal detection
 - Evidence image storage
 - Reduced manual monitoring
 - Fast alert generation
 - User-friendly dashboard
 - Low-cost implementation
-

Limitations

- Single camera support
 - Accuracy depends on image quality
 - Poor lighting may affect recognition
 - Requires trained dataset
 - Internet notification not available
-

Future Enhancements

- Deep Learning (CNN/FaceNet)
 - Multi-camera surveillance
 - Cloud database integration
 - Mobile App support
 - SMS/Email alerts
 - GPS location tracking
 - CCTV integration
 - Real-time police notification
-

Testing Results

Test Case	Expected Result	Status
Authorized Face	Access Granted	Pass
Criminal Face	Alert Generated	Pass
Unknown Face	Evidence Saved	Pass
Login Validation	Secure Access	Pass
Evidence Storage	Image Saved	Pass

CONCLUSION

SecureSight AI is an intelligent surveillance and criminal detection system developed using Artificial Intelligence, Computer Vision, and Face Recognition technologies. The system successfully detects human faces in real time, recognizes authorized individuals, identifies criminals from a predefined dataset, and generates alerts whenever suspicious activity is detected.

The project integrates OpenCV, LBPH Face Recognition, Flask, HTML, CSS, Bootstrap, and Python to provide a complete web-based surveillance platform. It automatically stores evidence images, maintains activity logs, and allows administrators to monitor security events through an interactive dashboard.

The successful implementation of this project demonstrates how Artificial Intelligence can improve security monitoring, reduce human effort, and enhance response time. SecureSight AI provides an efficient, cost-effective, and scalable solution for modern surveillance requirements and can serve as a foundation for future smart security systems.

References

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2. **OpenCV Documentation**
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4. **NumPy Documentation**
5. **GitHub Open Source Projects**
6. **Face Recognition Research Papers**
7. **Computer Vision Journals**